Course Name: Blockchain and its Applications (NOC26_CS34)

Assignment 2 - Week 2 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1

Alice employs the RSA cryptosystem with the prime numbers p=11 and q=17 to derive her public and private keys. Given that her public key is e=13, what is her corresponding private key d?

a) 71
b) 155
c) 37
d) 129

Answer: (c)

Detailed solution:
Compute $\phi(n)=(p-1)(q-1)=160.$
Find d as the modular inverse of $e=13$ modulo $\phi(n)=160$ using $d = e^{-1} \pmod{160}$, giving $d=37$

QUESTION 2

Bob receives a secure package from Alice, which was encrypted for **confidentiality** and digitally signed for **integrity**. To read the message and verify its origin, which keys must Bob use, and in what capacity?

a) Bob uses Alice's Public Key to decrypt the message and Bob's Private Key to verify the signature.
b) Bob uses Bob's Private Key to decrypt the message and Alice's Public Key to verify the signature.
c) Bob uses Bob's Public Key to decrypt the message and Alice's Private Key to verify the signature.
d) Bob uses Alice's Private Key to decrypt the message and Bob's Public Key to verify the signature.

Answer: (b)

Detailed solution:
For Confidentiality (Decryption): The message was originally encrypted by Alice using Bob's Public Key. Therefore, only Bob's Private Key can be used to decrypt and read it.

For Integrity/Authentication (Verification): The signature was created by Alice using Alice's Private Key. To verify this signature, Bob must use the corresponding Alice's Public Key.

QUESTION 3

Which statement best captures non-repudiation in blockchain transactions enabled by digital signatures?

a) Anyone can change a signed transaction without detection.
b) The sender can later deny authoring the transaction even if it's signed.
c) The sender cannot plausibly deny authoring the transaction, and others can verify the origin.
d) Digital signatures hide the transaction contents from all nodes.

Answer: (c)

Detailed solution:
A valid signature binds a transaction to the signer's private key, enabling verifiability and **non-repudiation**.

QUESTION 4

In a typical blockchain transaction flow, what does verifying Alice's signature using Alice's **public** key ensure?

a) The message is encrypted for Alice only
b) The message definitely remained confidential on the network
c) The transaction was authorized by Alice and was not altered after signing
d) The miners will include the transaction in the next block for sure

Answer: (c)

Detailed solution:
Signature verification checks **authenticity (who signed)** and **integrity (not modified)**, not confidentiality or inclusion guarantees.

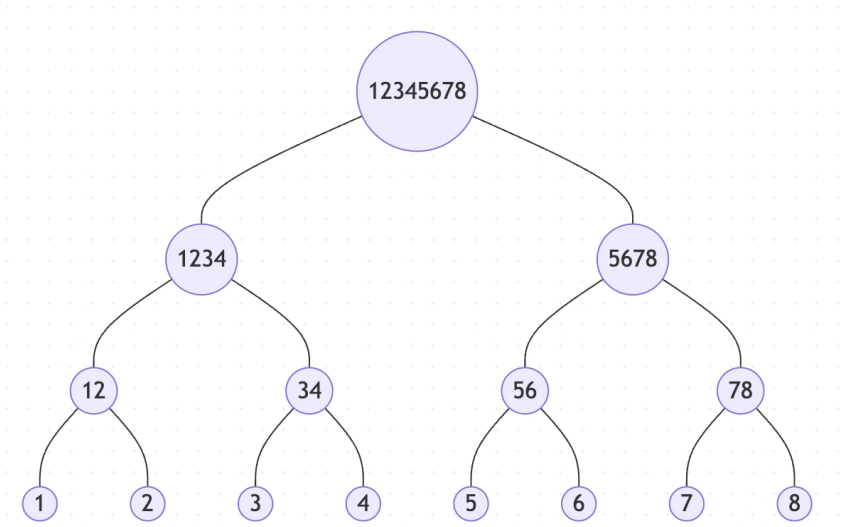
QUESTION 5

Consider 8 data points labeled 1 to 8. The post-order traversal of the Merkle Tree is provided as follows (where 1 represents the hash of data point 1, 56 denotes the combined hash of 5 and 6, and so on):

a) {12345678, 1234, 12, 1, 2, 34, 3, 4, 5678, 56, 5, 6, 78, 7, 8}
b) {1, 2, 12, 3, 4, 34, 1234, 5, 6, 56, 7, 8, 78, 5678, 12345678}
c) {1, 12, 2, 3, 34, 4, 1234, 5, 56, 6, 7, 78, 8, 5678, 12345678}
d) {12, 1, 2, 34, 3, 4, 1234, 56, 5, 6, 78, 7, 8, 5678, 12345678}

Answer: (b)

Detailed solution:



Post-order traversal: {1, 2, 12, 3, 4, 34, 1234, 5, 6, 56, 7, 8, 78, 5678, 12345678}

QUESTION 6

In the Bitcoin Proof of Work (PoW) system, which field in the block header do miners constantly modify to find a hash that meets the difficulty target?

e) Previous Block Hash
f) Merkle Root
g) Nonce
h) Timestamp

Answer: (c)

Detailed solution:
The **Nonce** is the field that miners change arbitrarily. They repeatedly hash the block header with different nonce values until the resulting hash falls below a specific target value.

QUESTION 7

Which option **most clearly conflicts** with the typical design goals of public cryptocurrencies?

a) Open participation with peer-to-peer propagation
b) Tamper-evident append-only ledger
c) Central authority that can unilaterally reverse valid transactions
d) Distributed validation through consensus

Answer: (c)

Detailed solution:
Public cryptocurrencies are designed to avoid a central party that can arbitrarily rewrite history.

QUESTION 8

Which of the following accurately describes the role of Miners in the Bitcoin network?

a) They are central authorities appointed by the Bitcoin Foundation to issue currency.
b) They use Proof of Stake (PoS) to vote on valid transactions based on their coin wealth.
c) They solve computational puzzles (Proof of Work) to validate transactions and secure the network in exchange for rewards.
d) They primarily focus on encrypting user passwords to prevent hacking.

Answer: (c)

Detailed solution:
Miners in Bitcoin operate on a Peer-to-Peer (P2P) network using Proof of Work (PoW). Their primary role is to bundle transactions into blocks and solve cryptographic puzzles. They are incentivized to do this work through block rewards and transaction fees.

QUESTION 9

In the context of the Bitcoin network, how do nodes resolve a temporary ‘fork’ (where two different valid blocks are found at roughly the same time)?

a) They choose the block that contains the most transactions.
b) They follow the "Longest Chain" rule (the chain with the most cumulative work).
c) A central server decides which block is valid.
d) Both blocks are discarded, and miners restart.

Answer: (b)

Detailed solution:
To maintain consensus without a central authority, Bitcoin nodes accept the chain that has the most cumulative Proof of Work (often referred to as the **longest chain**) as the valid truth.

QUESTION 10

In the context of blockchain distributed systems, what specific problem does the consensus algorithm (like Proof of Work) attempt to solve?

a) The Data Storage Problem: Ensuring all nodes have infinite storage capacity.
b) The Byzantine Generals Problem: Reaching agreement even when some participants may act maliciously or send conflicting information.
c) The Latency Problem: Ensuring data travels at the speed of light between nodes.
d) The Encryption Problem: Making sure no one can read the transaction details.

Answer: (b)

Detailed solution:
The core challenge in decentralized networks is the **Byzantine Generals Problem**. A robust consensus algorithm ensures that the honest nodes in the network can agree on a single history (ledger state) even if some nodes ("traitors") try to propagate false information or double-spend coins.